Few-Shot LLMs versus Domain-Adapted Encoders for Named Entity Recognition in Decisions from a Brazilian Court of Accounts

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Abstract. Decisions issued by Brazilian Courts of Accounts impose fines, obligations, reimbursements, and recommendations that staff transcribe into monitoring registries. We present `decicontas.br`, a public Named Entity Recognition (NER) dataset of 861 decisions of the Court of Accounts of the State of Rio Grande do Norte (TCE/RN) with four entity types—`MULTA` (fine), `OBRIGACAO` (obligation), `RESSARCIMENTO` (reimbursement), and `RECOMENDACAO` (recommendation)—and a comparative evaluation of nineteen extraction models. The dataset includes a confident-learning audit with paired releases and a three-annotator agreement study over the whole corpus (Fleiss $\kappa = 0.879$; pairwise span F1 0.824–0.838), with adjudication of the divergences. Nine few-shot LLMs are compared with ten supervised baselines, including seven legal- or government-domain encoders, all reading complete documents, under a shared span-scoring protocol (IoU ≥ 0.5) with paired bootstrap inference and Holm correction over all 171 model pairs. DeepSeek-V4-Flash achieves macro span F1 0.871, ahead of the strongest supervised baseline, LegalBert-pt (0.821), by $+0.049$ (95% CI $[-0.004, +0.103]$, $p = 0.070$, $p_{\text{Holm}} = 1.00$): the difference is not significant, and three encoders belong to the leader's group together with GPT-4.1 (0.823) and GPT-5.1 (0.811). Rescored against the first author's pre-adjudication annotation and the two independent annotations, the leader keeps a numerical lead of $+0.029$ to $+0.073$ over the best supervised model, significant only against Annotator 2. The ranking depends on the matching criterion: under exact span matching the encoders lead (0.686 against 0.421), so the two paradigms are equivalent for locating commands for human review, and the encoders are clearly better at automatic delimitation. Data and code: github.com/eduardoplima/decicontas.br.

Keywords. Annotated Legal Corpus, Named Entity Recognition, Large Language Models, Brazilian Portuguese, Legal Informatics, Courts of Accounts, Few-Shot Learning, Inter-Annotator Agreement, Label-Error Detection

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1. Introduction

1.1. Unstructured Decisions and Accountability Gaps in Courts of Accounts

Named Entity Recognition (NER) underpins document analysis and compliance monitoring in the legal domain, but faces specialized terminology, long sentences, and entity types absent from general-purpose inventories [1], which has motivated a family of legal corpora and domain-adapted models for Brazilian Portuguese [1,2,3]. Brazilian Courts of Accounts (*Tribunais de Contas*) exercise external control over public administration. Their collegiate decisions impose pecuniary sanctions, enforceable obligations, reimbursement orders, and non-binding recommendations, under a constitutional mandate that gives debt and fine rulings the force of enforceable titles (Art. 71, § 3 of the 1988 Federal Constitution [4]). These directives, however, are published as free text, and over 93% of Brazilian Courts of Accounts disseminate their determinations in fragmented and formalistic ways, hampering follow-up and accountability [5]. At the Court of Accounts of the State of Rio Grande do Norte (TCE/RN), the institution studied here, internal rules mandate a General Registry for Decision Monitoring (CGAD), whose subregistries of fines, refunds, and recommendations are still fed by clerks who read each decision and manually transcribe the relevant passages. In a 2024 survey, 60% of these courts reported AI-based solutions in use and 53% listed automating repetitive tasks as a strategic goal [6]. Manual transcription is error-prone, and every error propagates to the registry that grounds enforcement follow-up.

Related work on Courts of Accounts includes hybrid classification of Italian *Corte dei Conti* decisions [7] and information retrieval over Brazilian federal audit-court decisions [8]. Our focus is the span-level extraction of commands for institutional monitoring.

1.2. Objectives and Contributions

This paper evaluates extraction configurations for identifying and typing the decision elements that feed the CGAD. We contribute **a public dataset with an explicit quality protocol**: 861 decisions annotated with four audit entities, a confident-learning audit [9] with pre- and post-review releases, and independent re-annotation of the whole corpus by two domain experts. We provide **a comparative evaluation of nineteen models**: nine few-shot LLMs and ten supervised baselines, including seven legal- or government-domain encoders, all reading complete documents and scored under a shared span-level protocol with paired document-level bootstrap and Holm correction [10]. We also provide **a sensitivity analysis to reference annotation**, rescoring the same predictions against the pre-adjudication and the two independent annotations, alongside exact-match, model-selection, and prompting analyses and an error analysis linked to institutional review.

2. Related Work

2.1. NER Corpora for Brazilian Legal Text

LeNER-Br [1] was the first public NER corpus for Brazilian legal text; UlyssesNER-Br [2] shifted to the legislative domain; and CDJUR-BR [3] advanced annotation rigor over

a state court's procedural documents, with expert double annotation (Cohen's $\kappa = 0.69$) and adjudication. Closest to our setting, Menezes-Neto et al. introduced LexCare.br and evaluated few-shot LLMs for healthcare-related NER in Brazilian judicial decisions [11], whose prompting-based evaluation we extend with supervised baselines and statistical inference. Within the AI&Law community, Angelidis et al. [12] built an NER corpus and pipeline for Greek legislation, and Costa et al. [13] proposed a semantic annotation pipeline for Brazilian judicial decisions—the closest precedent in venue and material, though targeting ontology-driven annotation rather than a released NER benchmark. Beyond NER, LegalBench-BR [14] contributes a benchmark of Brazilian legal decision classification tasks for LLM evaluation. JurisTCU [8], 16,045 decisions of the federal Court of Accounts, is the closest audit-court resource, but it was built for information retrieval and carries no span annotation of the commands a decision issues.

2.2. Portuguese Legal and Governmental Encoders

BERTimbau [15] remains the reference Portuguese encoder, with a family of domain-adapted variants around it: JurisBERT and BERTimbauLaw on Brazilian legislation and case law [16], LegalBert-pt on documents from ten state courts [17], Legal-BERTimbau on European Portuguese supreme-court text [18], GovBERT-BR on governmental text [19], and RoBERTaLexPT [20].

2.3. LLMs for Named Entity Recognition

Few-shot prompting allows LLMs to extract from a handful of in-context demonstrations, without weight updates [21]. GPT-NER recasts NER as generation with markup tokens [22], and chain-of-thought prompting [23] adds intermediate reasoning, with mixed evidence in extraction tasks. In the legal domain, Lee et al. [24] applied few-shot learning to legal argument NER, and Adhikary et al. [25] used few-shot LLMs to extract case attributes for downstream prediction. Throughout we use *function calling*, in which the model fills the typed arguments of a client-declared function. Comparative studies in specialized domains report LLMs approaching or surpassing fine-tuned encoders in low-annotation regimes [11], but rarely control the prompting technique, quantify uncertainty, or test whether their conclusions survive a change of reference annotation.

3. The Decicontas.br Dataset

3.1. Source and Target Entities

The dataset comprises 861 decisions from collegiate sessions of the TCE/RN (Section 1.1), drawn from the institutional database. Four entity types mirror the CGAD subregistries [26].

`MULTA` (fine) is a pecuniary sanction for administrative or fiscal misconduct; `OBRIGACAO` (obligation) a binding duty imposed on the audited entity, including mandatory determinations; `RESSARCIMENTO` (reimbursement) a mandated reimbursement to the public treasury for established losses; and `RECOMENDACAO` (recommendation) a formally documented, non-binding recommendation addressed to an audited entity to improve its administrative controls (the guideline excludes recommendations addressed

to units of the Court itself). These are legal institutes with distinct enforcement regimes, not arbitrary NLP labels: fines and reimbursement rulings constitute enforceable titles [26], obligations bind under penalty of sanction, and recommendations feed future audit planning. Extraction errors therefore carry asymmetric legal costs—a missed fine may leave a sanction unregistered, while a spurious extraction may create an unfounded registry entry—which motivates both the strict annotation guidelines and the mandatory human review in deployment.

The corpus comprises 861 documents: 223 contain at least one entity, while the remaining 638 (74.1%) are true negatives — filings and clean-account judgments with no registrable command. All were read and validated; keeping them reflects the class prior of the production task, where a system must also learn *not* to extract. The corpus carries 429 entities: 209 `MULTA`, 122 `OBRIGACAO`, 66 `RESSARCIMENTO` and 32 `RECOMENDACAO`. Spans are long (per-class median lengths between 246 and 372 characters), which motivates the span-level evaluation with partial matching adopted in Section 4.2.

Documents are short and formulaic: 135.7 whitespace tokens on average, rising to 236.3 in the informative subset. Each class is anchored by strong lexical markers: the stem *ressarc-* appears within the first five tokens of 62 of the 66 `RESSARCIMENTO` spans (93.9%) and *recomend-* within the first five of 28 of the 32 `RECOMENDACAO` spans (87.5%), while the corresponding figures for `MULTA` (*mult-*, 171/209) and `OBRIGACAO` (*determin-*, 84/122) are markedly lower at 81.8% and 68.9%. These are frequent but individually insufficient cues: the same vocabulary occurs in non-dispositive passages that must *not* be extracted, where models overextract (Section 5.6).

3.2. Annotation and Guidelines

The reference annotation was produced by the first author, an External Control Auditor at the Rio Grande do Norte State Court of Accounts, in Label Studio under deliberately strict guidelines: spans must be literal copies of the source and inference is forbidden, because extracted passages are later re-located by string matching and feed official registries that admit no interpolated content. Tokenization is canonical whitespace splitting, shared by every producer and consumer of the released files.

3.3. Label-Error Audit with Confident Learning

Annotation errors were audited with confident learning [9], which estimates the joint distribution of observed and latent true labels from out-of-sample predicted probabilities and flags the examples whose observed label is most likely wrong. The probabilities came from an ensemble of two complementary token models (a linear classifier over Portuguese BERT embeddings and a fine-tuned Portuguese BERT token classifier). The audit flagged 794 suspicious token groups; the 567 groups with ensemble confidence ≥ 0.95 were manually reviewed one by one, and the remaining 227 kept their original labels. Among the 567 reviewed groups, 23 (4.1%) were changed and the remaining suggestions were rejected. This proportion describes the reviewed subset, not an estimated error rate for the entire corpus. Its flagging profile is dominated by suspected omissions—of the 4,650 flagged tokens, the largest blocks are `O` $\rightarrow$ `OBRIGACAO` (640) and `O` $\rightarrow$ `RECOMENDACAO` (547)—but most of these suggestions did not survive human review. Both the pre- and post-review versions are published as paired releases (JSON,

JSONL, CoNLL, and BRAT formats with checksums), so that any result can be reproduced against either gold standard; all experiments in this paper use the adjudicated version (release 1.3.0), in which the names of the natural persons who are parties to the proceedings are pseudonymised, one token per token. The corpus is maintained alongside the Court's human-in-the-loop registry pipeline and undergoes periodic revisions and corrections, which may change entity counts between releases; each release is recorded in the repository.

3.4. Inter-Annotator Agreement

To quantify the reliability of the schema and guidelines, the full corpus was independently re-annotated under a blind protocol by two staff members of the Court's unit responsible for feeding the CGAD—domain experts in exactly the task the schema formalizes—with the post-audit author annotation (A1) as the third version. Agreement is computed over all 861 documents at three granularities: token-level Cohen's κ per pair and Fleiss' κ overall [27], span-level pairwise F1 under the same matcher and the same token units as the model evaluation, and document-level presence. Token-level κ ranges from 0.864 to 0.899 with Fleiss $\kappa = 0.879$—"almost perfect" on the Landis–Koch scale [28]—and pairwise span F1 from 0.824 to 0.838 (Table 1). Class confusions are rare (13 cases across the three pairs, ten of them between the two pecuniary institutes `MULTA` and `RESSARCIMENTO`). Boundary divergences often reflect granularity: in document 95, for instance, the ruling imposes a single sanction of R$ 10,000.00 and then itemizes it into two R$ 5,000.00 parcels; A1 follows the ruling and annotates one `MULTA` span, whereas both independent annotators annotated one span per parcel. The dominant divergence is *registration*: for `RECOMENDACAO`, the two independent annotators recorded more recommendations than A1 (33 and 28 spans against 22; three of Annotator 3's address the Court's own units and fall outside the guideline), giving pairwise F1 of 0.764 and 0.760 against it but 0.820 between themselves, and document-presence κ of 0.704 and 0.693 against it but 0.778 between themselves. No matched span is confused between `OBRIGACAO` and `RECOMENDACAO`, suggesting that registrability decisions are a major source of disagreement. A robustness check against the pre-audit annotation (κ 0.853/0.866) also shows substantial token-level agreement.

The divergences were then adjudicated. Under the same matching, 167 items in 85 documents were listed: 125 spans marked by A2 and/or A3 but absent from A1 (40 by both), 35 A1 spans marked by neither, and 7 label disagreements. The first author decided each item in an interface showing the three annotations and never a model prediction: 33 spans were adopted (22 marked by both annotators) and 90 rejected (including the three addressed to the Court's own units); of the A1 spans, 14 were removed, 18 kept and one replaced; six label disagreements kept A1's label and one span was re-delimited; the four undecided items keep A1. The result is the gold used here: 429 entities ($+34$, -16), `RECOMENDACAO` rising from 22 to 32. Agreement is not measured against the adjudicated gold, which incorporates A2's and A3's spans.

Table 1. Inter-annotator agreement over the 861 documents. A1 is the first author's post-audit annotation, before adjudication; A2/A3 are independent expert annotators.

Pair	Token κ	Span F1 (micro)	Span F1 (macro)
A1 × A2	0.864	0.824	0.795
A1 × A3	0.875	0.828	0.792
A2 × A3	0.899	0.838	0.812
Fleiss κ (3-way)	0.879	—	—

4. Experimental Setup

4.1. Models

LLMs (few-shot). Nine instruction-tuned models: `gpt-4.1`, `gpt-4.1-mini`, `gpt-4.1-nano`, `gpt-5-mini`, `gpt-5.1` and `gpt-5.2` (OpenAI), the open-weights `deepseek-v4-flash` [29] (MIT), `llama-3.3-70b` (Meta) and `qwen2.5-72b` (Alibaba). All were consumed through managed APIs rather than self-hosted: the OpenAI models and DeepSeek via Azure AI Foundry (Brazil South), the other two via OpenRouter. Dated snapshots were pinned where exposed. Decoding used temperature 0, top-p 1, a 4096-token output limit and seed 1007 on Azure, where a seed is honoured on a best-effort basis and does not guarantee bit-identical output; OpenRouter exposes no seed, and the three `gpt-5.x` are reasoning models whose API ignores sampling parameters. Structured output was obtained via function calling against a typed schema mirroring the registry: one list per entity class, each item carrying a description field whose value must be a literal excerpt of the decision. The prompt embeds twelve curated exemplars, four with empty expected output, teaching the model not to extract from non-registrable commands. The prompt was frozen and shared across models in the main comparison. Beyond static few-shot, two techniques are evaluated on five representative LLMs: *chain-of-thought* [23] and a *two-stage* variant that first lists candidate passages and then structures them in a second call.

Supervised baselines. Ten models trained with 5-fold cross-validation: a BiLSTM-CRF [30], the generic BERTimbau base and large [15], and seven domain-adapted encoders [18,16,17,31,32,19]. Encoders read each document in windows of 512 subwords overlapping by 320; a word covered by two windows takes the prediction of the window in which it lies farthest from an edge. The BiLSTM-CRF accepts 1,100 tokens, above the longest document (1,056), so no model truncates its input. Out-of-fold predictions feed the comparison on the same 861 documents. For each model, the configuration (encoders: learning rate $\{2,3,5\} \times 10^{-5}$ and warmup ratio $\{0,0.1\}$; BiLSTM-CRF: eight combinations of hidden size, dropout and learning rate) was selected by span micro F1 on a single 80/20 development split, and checkpoints by strict IOB2 F1.

4.2. Metrics and Statistical Protocol

We report token-level F1 over collapsed labels and span-level precision, recall, and F1 in micro and macro aggregations. Adapting entity-level partial-matching evaluation [33], a prediction matches a gold entity of the same class when their IoU is ≥ 0.5; greedy bipartite matching uses each prediction and each entity at most once. **All predictions are**

scored in one unit: spans are projected onto canonical whitespace token indices before matching, and a token belongs to a span whenever the two overlap, using the same rule as the gold BIO conversion. Every model is scored against the full gold of every document. A deterministic filter, applied identically to every model's predictions before scoring, discards predictions on deliberations addressed to units of the Court itself; no reference entity overlaps these clauses, and no discarded prediction overlapped one.

LLM outputs are strings, so they are first re-located in the decision. A string that occurs literally is placed at that occurrence; only non-literal strings fall back to fuzzy search, a 500-character window sliding by 100 characters and scored by normalized Levenshtein partial ratio (≥ 80). For the leader, 395 of 479 strings (82.5%) are literal copies and none fails to align; failures are zero for six of the nine LLMs and peak at 1.4% for GPT-4.1-nano. Strings that cannot be located are discarded before scoring, so the metrics describe validated output.

Macro span F1 is the primary metric: it assigns equal weight to the four entity classes. Uncertainty is quantified by paired document-level bootstrap (10,000 resamples, 95% percentile CIs, two-sided p-values); reported differences are observed differences computed before rounding. Holm correction [10] is applied over all 171 model pairs. A non-significant difference is not evidence of equivalence.

Two asymmetries remain. First, the supervised configuration was selected once per model on the global 80/20 split, *not* within each cross-validation training fold, so the 173 development documents reappear in evaluation (20.1% of each test fold on average); Section 5.3 rescores every model without them. Second, exposure to negatives differs: about three quarters of the documents the encoders train on are negative (74.1% of the corpus), whereas only four of the twelve prompt exemplars are; the ratio in the prompt was not varied.

5. Results and Discussion

5.1. Overall Performance

The strongest few-shot LLM leads numerically, without a significant margin over the best encoders. DeepSeek-V4-Flash (0.871; Table 2) exceeds the strongest supervised model, LegalBert-pt (0.821), by $+0.049$ (95% CI $[-0.004, +0.103]$; $p = 0.070$, $p_{\text{Holm}} = 1.00$), BERTimbau-large (0.813) by $+0.058$ ($[+0.001, +0.114]$; $p = 0.045$, $p_{\text{Holm}} = 1.00$) and BERTimbau-base (0.778) by $+0.094$ ($[+0.032, +0.160]$; $p = 0.002$, $p_{\text{Holm}} = 0.126$); only the BiLSTM-CRF (0.621; $+0.251$, $p_{\text{Holm}} < 0.001$) and the remaining encoders are significantly behind. The gap is larger for macro than for micro averaging ($+0.043$; LegalBert-pt is second overall on micro F1, ahead of GPT-4.1) and is concentrated on `RECOMENDACAO` (about 0.029 of the 0.049).

An open-weights model leads, without a significant difference from GPT-4.1 or GPT-5.1. DeepSeek-V4-Flash exceeds GPT-4.1 (0.823) by $+0.048$ (95% CI $[+0.016, +0.080]$, $p = 0.002$), a difference that does not survive Holm correction ($p_{\text{Holm}} = 0.149$); GPT-5.1 (0.811; $p_{\text{Holm}} = 0.071$) likewise remains in the leader's group, which thus contains three LLMs and three encoders. Of the 171 model pairs, 124 differ at the nominal 5% level and 100 after Holm correction.

The top supervised models are not significantly different from one another. LegalBert-pt (0.821), BERTimbau-large (0.813) and BERTimbau-base (0.778) lead

Table 2. Overall results on `decicontas.br` (861 documents, 429 entities). Span metrics use IoU ≥ 0.5 over whitespace token indices; P and R are span precision and recall. Sorted by macro span F1. †Leader's group: not distinguishable from DeepSeek-V4-Flash after Holm correction over the 171 pairs.

Model	Type	Token F1	P	R	$F1_\mu$	$F1_M$
DeepSeek-V4-Flash†	LLM	**0.914**	**0.884**	0.872	**0.878**	**0.871**
GPT-4.1†	LLM	0.886	0.760	0.893	0.821	0.823
LegalBert-pt†	Sup.	0.882	0.843	0.828	0.835	0.821
BERTimbau-large†	Sup.	0.874	0.832	0.809	0.820	0.813
GPT-5.1†	LLM	0.859	0.728	0.886	0.799	0.811
BERTimbau-base†	Sup.	0.869	0.831	0.793	0.811	0.778
GPT-4.1-mini	LLM	0.865	0.686	0.907	0.781	0.771
GPT-5.2	LLM	0.834	0.603	0.907	0.724	0.755
LegalBERTPT-br	Sup.	0.813	0.786	0.718	0.750	0.702
JurisBERT	Sup.	0.813	0.776	0.720	0.747	0.681
Legal-BERTimbau-base	Sup.	0.857	0.781	0.772	0.776	0.671
Qwen2.5-72B	LLM	0.792	0.637	0.821	0.717	0.664
BERTimbauLaw	Sup.	0.825	0.869	0.725	0.790	0.643
GPT-5-mini	LLM	0.624	0.314	**0.921**	0.469	0.629
Legal-BERT-STF	Sup.	0.812	0.818	0.704	0.757	0.625
BiLSTM-CRF	Sup.	0.830	0.828	0.639	0.721	0.621
GovBERT-BR	Sup.	0.774	0.853	0.662	0.745	0.585
GPT-4.1-nano	LLM	0.643	0.411	0.639	0.500	0.451
Llama-3.3-70B	LLM	0.425	0.673	0.249	0.364	0.267

the supervised group; LegalBert-pt's differences from BERTimbau-large ($+0.009$, $[-0.038, +0.056]$) and BERTimbau-base ($+0.045$, $[-0.001, +0.096]$) have intervals containing zero; 19 of the 45 supervised pairs survive the correction, each opposing one of these three to one of the other seven. The generic BERTimbau-base is third of ten, above the band of the remaining domain-adapted encoders (0.585–0.702). No fold yields $F1 = 0$, and per-fold standard deviations span 0.036–0.094. Supervised models favor precision (0.776–0.869) over recall (0.639–0.828), GPT-4.1 and GPT-5.1 favor recall, and the leader is balanced (0.884/0.872). This profile motivates evaluating LLMs as pre-annotators, although the relative effort of correcting insertions and omissions was not measured.

Aggregation changes the ranking: GPT-5-mini is second worst among LLMs on micro span F1 (0.469), with precision 0.314, yet its macro score (0.629) exceeds three supervised models. Llama-3.3-70B returns empty lists for 753 of 861 documents. The three `gpt-5.x` models rank below GPT-4.1 (GPT-4.1 exceeds GPT-5.2 by $+0.068$, $p < 0.001$, $p_{\text{Holm}} < 0.001$).

5.2. Performance per Entity Type

On `MULTA`, the majority class, span F1 is 0.907 for DeepSeek-V4-Flash, 0.865 for GPT-4.1, 0.828 for BERTimbau-large and 0.861 for LegalBert-pt; on `RESSARCIMENTO` it is 0.823, 0.779, 0.759 and 0.824 (LegalBert-pt level with the leader). On `OBRIGACAO`, DeepSeek-V4-Flash (0.849) and BERTimbau-large (0.842) are level, with LegalBert-pt at 0.813 and GPT-4.1 at 0.751. The separation is largest on `RECOMENDACAO`, the rarest class (32 instances): the leader's-group LLMs reach 0.896–0.906, against 0.821 for BERTimbau-large (23 of 32), 0.787 for LegalBert-pt (24 of 32) and 0.667 for BERTimbau-base; the other seven supervised models stay below 0.52. It concentrates most of the macro gap and the greatest human disagreement (Section 3.4).

Table 3. The central contrast under each reference: the adjudicated gold, the first author's pre-adjudication annotation (A1) and the two independent annotations. Observed differences with paired bootstrap 95% CIs; Spearman correlation of the full ranking against the gold's.

Reference	DeepSeek-V4-Flash	Best supervised	Gap	95% CI	Spearman
Gold	0.871	0.821 (LegalBert-pt)	$+0.049$	$[-0.004, +0.103]$	1.00
A1	0.799	0.769 (BERTimbau-large)	$+0.029$	$[-0.027, +0.083]$	0.99
A2	0.873	0.800 (LegalBert-pt)	$+0.073$	$[+0.019, +0.124]$	0.99
A3	0.817	0.774 (BERTimbau-large)	$+0.043$	$[-0.015, +0.104]$	0.97

5.3. Robustness: Matching Criterion, Model Selection, and Human Agreement

Four analyses assess sensitivity to evaluation and selection choices. First, *matching criterion*: IoU thresholds of 0.3 and 0.7 preserve the macro ranking (Spearman 0.983 and 0.993), whereas exact matching reorders it (Spearman 0.397). Under exact matching the encoders lead: LegalBert-pt has macro span F1 0.686 (micro 0.711) against 0.421 (0.392) for the leader, and all ten supervised models exceed every LLM on macro F1.

LLM boundary errors are systematic and local. Of the leader's 374 matched spans, only 29 reproduce the gold character offsets exactly (exact micro F1 0.068 in characters against 0.392 in tokens): 208 end one character early (133 omit a final semicolon, 75 a full stop), which the token projection absorbs, and 74 start thirteen characters early by including headings such as *RENOVAÇÃO DA*. Partial matching therefore fits a deployment in which extracted passages are candidates for human review; automatic registry population needs the encoders' boundary accuracy.

Second, *model selection*: rescoring every model on the 688 documents never used for configuration selection lowers all scores slightly (LegalBert-pt 0.796, BERTimbau-large 0.784, the leader 0.852); the central gap is $+0.056$ (95% CI $[-0.011, +0.124]$). Third, restricting evaluation to the 223 documents with entities raises the leader by 0.005, the other leading LLMs by 0.037–0.053 and the supervised models by at most 0.012, because LLM false positives on empty decisions weigh more. Fourth, the leader's micro and macro span F1 (0.878; 0.871) exceed the human pairwise ranges (0.824–0.838; 0.792–0.812), measured between annotators, not against the gold. Human agreement is an empirical reference, not a ceiling.

5.4. Sensitivity to the Choice of Reference

The gold reflects adjudication decisions taken by the first author, and Section 3.4 shows the two experts diverge from A1 on `RECOMENDACAO`, so a conclusion measured against the gold could be measuring adherence to its annotation policy rather than extraction quality. We test this by rescoring *the same predictions* against A1, A2 and A3 (Table 3).

Rankings remain similar, and the same LLM leads under all four references. The gap keeps its sign, and its interval excludes zero only against A2 ($p = 0.009$). The variation is concentrated on `RECOMENDACAO`: the leader's F1 on this class is 0.769 (A1), 0.903 (gold), 0.921 (A2) and 0.873 (A3), because the gold and the two annotators register more recommendations than A1 (32, 33 and 25 spans under the scope guideline against 22); LegalBert-pt's goes from 0.667 (A1) to 0.787 (gold), 0.806 (A2) and 0.741 (A3). After adjudication, the gold's registration threshold lies between A1's and A2's. This is descriptive robustness on the same corpus, not a test of cross-court generalization.

5.5. Prompting Technique

Chain-of-thought [23] reduces the leader's macro span F1 by 0.106 (0.871 $\rightarrow$ 0.765), a loss of precision (0.884 $\rightarrow$ 0.766), and raises GPT-4.1 (0.823 $\rightarrow$ 0.831), GPT-5.2 (0.755 $\rightarrow$ 0.811) and Qwen2.5-72B (0.664 $\rightarrow$ 0.680). A two-stage variant changes the leaders little (DeepSeek-V4-Flash 0.862, GPT-4.1 0.835) and lowers Llama-3.3-70B to 0.136.

5.6. Error Analysis

For the leader, 374 of the 429 gold entities are matched with the correct label (recall 0.872), and 374 of its 423 scored predictions are correct (precision 0.884). Of the 55 missed entities, 26 receive no overlapping prediction and 29 only by predictions that fail the match. Of the 49 incorrect predictions, 8 overlap no gold entity, 37 overlap one below the threshold and 4 match it with the wrong class.

Unmatched insertions fall mostly on `OBRIGACAO` (6 of the 8; 1 `MULTA`, 1 `RECOMENDACAO`), and none is fabricated. Class confusion accounts for only 4 of the 104 errors.

5.7. Limitations

The corpus comes from one court and is concentrated in one year. The gold reflects the judgment of one adjudicator, who is also the first annotator; rescoring against the pre-adjudication and the two independent annotations preserves the leading model. Configuration selection is global rather than nested; rescoring without the development documents moves the central gap by less than 0.01, but nested cross-validation was not run. Varying the positive/negative ratio of the prompt is left for future work, as are per-model prompt optimization and the evaluation of RoBERTaLexPT [20] and Sabiá-4 [34]. At Azure list prices, the leader processes the corpus for US$1.94 (US$0.35 with prompt caching), at a median 2.7 s per decision over a 50-document sample, against 0.04 s for LegalBert-pt on an Apple-silicon laptop; reviewer effort and productivity gains were not measured.

6. Conclusion

We introduced `decicontas.br`, a NER dataset of Brazilian state Court of Accounts decisions, with paired pre-/post-review releases, a confident-learning audit, and a three-annotator agreement study with adjudication. Under IoU $\geq$ 0.5, DeepSeek-V4-Flash leads with macro span F1 0.871, but its margin over the strongest supervised baseline, LegalBert-pt (+0.049), is not significant after Holm correction, and three encoders share the leader's group with GPT-4.1 and GPT-5.1. Its lead persists under the three alternative references, significantly only against one. Exact matching puts the encoders clearly ahead. Few-shot LLMs and domain-adapted encoders are therefore equivalent options for proposing candidate passages for human review, whereas automatic registry population requires a boundary fidelity that only the encoders approach.

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References

[1] de Araujo PHL, de Campos TE, de Oliveira RRR, Stauffer M, Couto S, Bermejo P. LeNER-Br: A dataset for named entity recognition in Brazilian legal text. In: Computational Processing of the Portuguese Language (PROPOR). vol. 11122 of LNCS. Springer; 2018. p. 313-23. doi:10.1007/978-3-319-99722-3_32.

[2] Albuquerque HO, et al. UlyssesNER-Br: A corpus of Brazilian legislative documents for named entity recognition. In: Computational Processing of the Portuguese Language (PROPOR). vol. 13208 of LNCS. Springer; 2022. p. 3-14. doi:10.1007/978-3-030-98305-5_1.

[3] Brito M, Pinheiro V, Furtado V, Monteiro Neto J, Bomfim FCJ, Costa ACF, et al. CDJUR-BR: A golden collection of legal documents from Brazilian justice with fine-grained named entities; 2023. arXiv preprint arXiv:2305.18315.

[4] Brasil. Constituição da República Federativa do Brasil de 1988; 1988. Diário Oficial da União. Available from: `https://www.planalto.gov.br/ccivil_03/constituicao/constituicao.htm`.

[5] Santiago LG, Goes AS, Petean GH, Bevilacqua S. A protocolar transparência das determinações dos Tribunais de Contas brasileiros: uma oportunidade de aprimoramento da accountability. Revista Controle. 2024;23(1). doi:10.32586/rcda.v23i1.947.

[6] Instituto Rui Barbosa, Atricon. Inteligência Artificial nos Tribunais de Contas: Avanços e Desafios. Brasília: IRB; 2024.

[7] Decker L, Palmirani M, Sapienza S, Pistone V. Hybrid classification of audit court decisions using online context-driven neural networks. CEUR Workshop Proceedings. 2023;3637.

[8] Fernandes LC, Ribeiro LS, de Castro MVB, Pacheco LAS, Sandes EFO. JurisTCU: A Brazilian Portuguese information retrieval dataset with query relevance judgments. Language Resources and Evaluation. 2026;60. doi:10.1007/s10579-025-09881-w.

[9] Northcutt CG, Jiang L, Chuang IL. Confident learning: Estimating uncertainty in dataset labels. Journal of Artificial Intelligence Research. 2021;70:1373-411. doi:10.1613/jair.1.12125.

[10] Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics. 1979;6:65-70.

[11] de Menezes-Neto EJ, Bezerra FLdO, Clementino MBM. Evaluating LLMs for healthcare-related named entity recognition in Brazilian judicial decisions. In: 2025 IEEE Symposium on Computers and Communications (ISCC); 2025. p. 1-6. doi:10.1109/ISCC65549.2025.11326242.

[12] Angelidis I, Chalkidis I, Koubarakis M. Named Entity Recognition, Linking and Generation for Greek Legislation. In: Legal Knowledge and Information Systems: JURIX 2018. vol. 313 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2018. p. 1-10. doi:10.3233/978-1-61499-935-5-1.

[13] Costa MZ, Robson DF, Vieira TBP, Bourguet JR, Guizzardi G, Almeida JPA. Automated Semantic Annotation Pipeline for Brazilian Judicial Decisions. In: Legal Knowledge and Information Systems: JURIX 2024. vol. 395 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2024. p. 226-38. doi:10.3233/FAIA241248.

[14] de Carvalho Neto PB. LegalBench-BR: A benchmark for evaluating large language models on Brazilian legal decision classification; 2026. arXiv preprint arXiv:2604.18878.

[15] Souza F, Nogueira R, Lotufo R. BERTimbau: Pretrained BERT models for Brazilian Portuguese. In: Intelligent Systems (BRACIS 2020). vol. 12319 of Lecture Notes in Computer Science. Cham: Springer; 2020. p. 403-17. doi:10.1007/978-3-030-61377-8_28.

[16] Viegas CFO, Costa BC, Ishii RP. JurisBERT: A new approach that converts a classification corpus into an STS one. In: Computational Science and Its Applications – ICCSA 2023. vol. 13956 of LNCS. Springer; 2023. p. 349-65. doi:10.1007/978-3-031-36805-9_24.

[17] Silveira R, Ponte C, Almeida V, Pinheiro V, Furtado V. LegalBert-pt: A pretrained language model for the Brazilian Portuguese legal domain. In: Intelligent Systems (BRACIS 2023). vol. 14197 of LNCS. Springer; 2023. p. 268-82. doi:10.1007/978-3-031-45392-2_18.

[18] Melo R, Santos PA, Dias J. A semantic search system for the Supremo Tribunal de Justiça. In: Progress in Artificial Intelligence (EPIA 2023). Springer; 2023. p. 142-54. doi:10.1007/978-3-031-49011-8_12.

[19] Silva MO, Oliveira GP, Costa LGL, Pappa GL. GovBERT-BR: A BERT-based language model for Brazilian Portuguese governmental data. In: Intelligent Systems (BRACIS 2024). vol. 15413 of LNCS. Springer; 2025. p. 19-32. doi:10.1007/978-3-031-79032-4_2.

[20] Garcia E, et al. RoBERTaLexPT: A legal RoBERTa model pretrained with deduplication for Portuguese. In: Proceedings of PROPOR 2024. ACL; 2024. p. 374-83.

[21] Brown TB, et al. Language models are few-shot learners. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 33; 2020. p. 1877-901.

[22] Wang S, et al. GPT-NER: Named entity recognition via large language models; 2023. arXiv preprint arXiv:2304.10428.

[23] Wei J, et al. Chain-of-thought prompting elicits reasoning in large language models; 2022. arXiv preprint arXiv:2201.11903.

[24] Lee SM, Tan YH, Yu HT. LeArNER: Few-shot Legal Argument Named Entity Recognition. In: Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law (ICAIL). ACM; 2023. p. 422-6. doi:10.1145/3594536.3595144.

[25] Adhikary S, Sen P, Roy D, Ghosh K. A case study for automated attribute extraction from legal documents using large language models. Artificial Intelligence and Law. 2024. doi:10.1007/s10506-024-09425-7.

[26] Tribunal de Contas do Estado do RN. Lei Complementar Estadual nº 464, de 5 de janeiro de 2012 (Lei Orgânica do TCE/RN); 2012. Available from: `https://www.tce.rn.gov.br/as/download/Legislacao/LCE_464.2012__Lei_Org%C3%A2nica.pdf`.

[27] Fleiss JL. Measuring nominal scale agreement among many raters. Psychological Bulletin. 1971;76:378-82. doi:10.1037/h0031619.

[28] Landis JR, Koch GG. The measurement of observer agreement for categorical data. Biometrics. 1977;33:159-74. doi:10.2307/2529310.

[29] DeepSeek-AI. DeepSeek-V4: Towards highly efficient million-token context intelligence; 2026. arXiv preprint arXiv:2606.19348.

[30] Lample G, Ballesteros M, Subramanian S, Kawakami K, Dyer C. Neural architectures for named entity recognition. In: Proceedings of NAACL-HLT; 2016. p. 260-70. doi:10.18653/v1/N16-1030.

[31] Ulysses Project, Brazilian Chamber of Deputies. LegalBERTPT-br: Sentence Embeddings Trained on Brazilian Legislative Texts; 2022. Hugging Face Model Hub. Available from: `https://huggingface.co/ulysses-camara/legal-bert-pt-br`.

[32] Domingues M. Language Model in the Legal Domain in Portuguese (Legal-BERT-STF); 2022. Hugging Face Model Hub. Available from: `https://huggingface.co/dominguesm/legal-bert-base-cased-ptbr`.

[33] Segura-Bedmar I, Martínez P, Herrero-Zazo M. SemEval-2013 Task 9: Extraction of Drug-Drug Interactions from Biomedical Texts (DDIExtraction 2013). In: Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013). Atlanta, Georgia, USA: Association for Computational Linguistics; 2013. p. 341-50. Available from: `https://aclanthology.org/S13-2056`.

[34] Laitz T, Almeida TS, Abonizio H, Malaquias Junior R, Bonás GK, Piau M, et al. Sabiá-4 Technical Report; 2026. arXiv preprint arXiv:2603.10213.